# Rhetorical Essay 2: Deconstructing of the Human Subject


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Humans were commonly seen as superior to animals due to their abilities as makers and superior minds. This was a very prominent idea in Mumford's reading discussing how humans are both *Homo Faber* and *Homo Sapiens* (tool maker and mind maker respectively)[1]. However, advancements in technology and computational theories have dismantled traditional theories of the human self, replacing notions of a soul, inner dialogue, and free will with a more technical definition. This stems from no longer viewing humans as an autonomous person, rather a being defined by their external behaviour and aactions as data points. This essay will look at contemporary notions of humans and individuals that blur the line between a human individual and a small part of a larger system, as well as the reduction of a human to an info-computationalist system rather than a unique autonomous self.

One of the first instances where the boundary between humans and machines were blurred was during WWII. In Peter Galison's "The Ontology of the Enemy", he described an elaborate process to create antiaircraft predictors. In creating these machines, they started to view the enemy human pilots as a "servo-mechanism" which helped determine the path of the aircraft. Galison quoted Wiener's "Summary Report for Demonstration" where Wiener noted that "the pilot behaves like a servo-mechanism, attempting to overcome the intrinsic lag due to the dynamics of his plane as a physical system"[2]. Here, Wiener choose to view the pilot as a mere part in a system, designed to correct and guide the aircraft, rather than a human with fully autonomous capabilities. Here, the enemy pilot was no more than a small mechanical part in this complex aircraft system, assigned the role of correcting error and guiding the plane like how the engine is assigned to power the aircraft. This assignment wasn't some grand philosophical breakthrough, rather a means to make the difficult notions of a human more practical. The complex notions of human autonomy and free will was nothing but a mere military problem, something to be overcome through engineering a solution to predict the complex mechanical system. As such, viewing humans as a mere cog in a machine is a much easier computation than to try to understand the complex physiological and psychological components that make us humans. This is how Galison framed Wiener's work as well as how Wiener justified this reduction. In fact, Wiener comes to terms with these complexities and states that "It does not seem even remotely possible to eliminate the human element as far as it shows itself in enemy behavior. Therefore, in order to obtain as complete a mathematical treatment as possible of the over-all control problem, it is necessary to assimilate the different parts of the system to a single basis, either human or mechanical. Since our understanding of the mechanical elements of gun pointing appeared to us to be far ahead of our psychological understanding, we chose to try to find a mechanical analogue of the gun pointer and the airplane pilot"[3]. This

[1] Mumford, Lewis. "Technics and the Nature of Man." Technology and Culture 7, no. 3 (Summer 1966): 309.

[2] Galison, Peter. "The Ontology of the Enemy: Norbert Wiener and the Cybernetic Vision." Critical Inquiry 21, no. 1 (Autumn 1994): 236

[3] Galison, "The Ontology of the Enemy", 241

demonstrates how Wiener viewed the reduction of the enemy pilot to a simple mechanical piece seemed necessary to him. To Wiener, viewing the enemy pilots as human made these systems messy and complicated beyong the current understanding of human psychology. As such, the complexities were discarded in a simpler view of humans as a mechanism in a larger machine.

The reduction of humans to a small part or mechnaism within a greater system, while problematic and arguably unethical, allowed for a new means of looking at individuals. Mumford coined the term of "Megamachine"[4] which described these complex systems comprised of human parts which followed the will of those in charge. These systems were designed to simplify the human into a small part of the system as a means to improve effeciency and create things no single part could create by itself. Things such as the great pyramids in Egypt were constructed by these large machines of human parts, ellaborate systems and rules to make each worker nothing but a simplistic function within the system. Mumford brought up more contemporary examples of the "Megamachine" such as large coorperations and enterprises or even military machines. Here all of these "Megamachines" simply follow one goal in mind with each human part doing their own function for the goal. In the case of military, this goes back to Wiener's research and reducing a pilot to a part of the plane machine. By reducing the human pilot to a mere part in a system, Wiener found ways of predicting the trajectory of a plane based on the past trajectory of that plane system. According to Galison, this servomechanical enemy and view of the human pilots became "the prototype for human physiology and, ultimately, for all of human nature"[5]. This reduction of human behaviour and actions made way for the "black-box" model which ignored the how or why of the function, but simply aimed to understand what it did based on the input. The notions of the "black-box" model were then applied to human psychology and Galison quotes an analysis of Wiener's work where Wiener argues "their equipment is probably one of the closest mechanical approaches ever made to physiological behavior. Parenthetically, the Wiener predictor is based on good behavioristic ideas, since it tries to predict the future actions of an organism not by studying the structure of the organism but by studying the past behavior of the organism"[6]. This completely ignores any sort of characteristics of what makes a human unique, but rather looks at the human as a merely predictable function of inputs and outputs. This solidifies the dismantlized nature of human to a mere log of their past actions and disrcards any sort of imperfections and blemishes that makes the subject human.

The idea that humans are mere data points in a complex system is akin to Deleuze's notions of humans as a dividual, disected into a "bank" of data. To introduce this reduction, Deleuze first introduces the concept of codes in societies of control which devalue the importance of a signature because these codes "mark access to information, or reject it"[7]. For these societies, they reduce individuals to essentially a code/password which permits an individual's access (or exclusion) to a system or space. This allows society to monitor and modulate an individual's

[4] Mumford, "Technics and the Nature of Man", 312

[5] Galison, "The Ontology of the Enemy", 233

[6] Galison, "The Ontology of the Enemy", 244

[7] Deleuze, Gilles. "Postscript on the Societies of Control." October 53 (Summer 1990): 5.

position and power at any time throughout the day. Think for example our CaseId's which allow certain people access to specific buildings at specific hours and can be activated and deactivated with a short script to a database. This sort of concept leads to Deleuze's idea where societies no "longer find [them]selves dealing with the mass/individual pair. Individuals have become 'dividuals,' and masses, samples, data, markets, or 'banks' "[8]. The etymology of the word "individual" defines itself as one that is a sum of indivisible/inseparable parts, but here Deleuze is using the term "dividual" to define a single being, implying that an individual human is being dismantled into separate data points. According to Deleuze, for these societies an individual is nothing more than a sample from a dataset used to run some sort of algorithm, a "bank" of data to be mined. Here, similar to how Wiener viewed humans as a servo-mechanism which produced data points and could be predicted, the society of control views each human individual as a unrefined data source that needs to be collected and processed. Deleuze gives the example of a hospital system where "the new medicine 'without doctor or patient' that singles out potential sick people and subjects at risk, which in no way attests to individuation—as they say—but substitutes for the individual or numerical body the code of a 'dividual' material to be controlled"[9]. Here, there is no more doctor and patient relationship but each individual is analyzed as a data point in a system that identifies and isolates subjects at risk. This is all done by looking at individuals not as humans but a combination of divisible parts, genetic history, diet, lifestyle choices etc. This is a direct example of the "dividual" concept in action where the human subject is dismantled into parts and used to complete their data profile. One's dismantled data profile becomes more important than the experiences that make the individual a whole.

This dismantling of the human can be viewed as either a natural process or a problematic ideology. For Turing, "Computation is not just a language of nature; it is the way nature behaves. Computing differs from mathematics in that computers not only calculate numbers, but more importantly they can produce real time physical behaviours"[10]. This shows how from Turing's perspective, all of nature is some form of computations and calculations that determine all sorts of physical behaviours. For Turing, human behaviours and actions being reduced to computational data points is natural because he believes everything in nature is derived from some sort of computation. As such, Turing believes that the reduction of a human to an info-computationalist system is both natural and logical because we as humans are part of nature and are subject to Turing's natural philosophy of being bound to a system of computation. Dodig-Crnkovic also analyzed Edward Fredkin who asserted that "humans are software running on a universal computer"[11] as well as Wolfram who believed "the complexity of behaviors and structures found in nature are derived (generated) from a few basic mechanisms. Natural phenomena are thus the products of computation processes"[12]. These minds believed that fundamentally the universe was a discrete system that could be mod-

[8]Deleuze, "Postscript", 5
[9]Deleuze, "Postscript", 7
[10]Dodig-Crnkovic, Gordana. "Alan Turing's Legacy: Info-computational Philosophy of Nature." In Computing Nature, edited by G. Dodig-Crnkovic and R. Giovagnoli, 115. SAPERE 7. Berlin: Springer-Verlag, 2013.
[11]Dodig-Crnkovic, "Alan Turing's Legacy", 117
[12]Dodig-Crnkovic, "Alan Turing's Legacy", 117

eled by a computer and set of computations (albeit a complex one). To the minds of the info-computationalists, humans, like nature, can be modeled by some set of computations and rules. There were even notions of cellular automata, which acted as a universal computational building block for explaining behaviours within living organisms. The tricky part about these cellular automata was not that they were perfect but they were a surprisingly accurate model. Thus, rather than simplifying the human to a data point, it is a systematic dismantling of human processes down to computational models and structured automata / systems.

In conclusion, the traditional view of the autonomous human subject has been systematically dismantled by technological and computational perspectives. From Wiener's servomechanical pilot and Mumford's notions of a Megamchine, to Deleuze's data-driven 'dividual,' the human has been re-conceptualized as a predictable and controllable component within a larger system. This reframing exchanges the complexities of the individual for a more practical, albeit reductive, model based on external behavior and data points. Ultimately, info-computationalism naturalizes this process, suggesting this deconstruction is not a mere simplification but an accurate reflection of a universe that is fundamentally governed by computation.

## Works Cited


Deleuze, Gilles. “Postscript on the Societies of Control.” October 53 (Summer 1990): 3–7.

Dodig-Crnkovic, Gordana. “Alan Turing’s Legacy: Info-computational Philosophy of Nature.” In Computing Nature, edited by G. Dodig-Crnkovic and R. Giovagnoli, 115–123. SAPERE 7. Berlin: Springer-Verlag, 2013.

Galison, Peter. “The Ontology of the Enemy: Norbert Wiener and the Cybernetic Vision.” Critical Inquiry 21, no. 1 (Autumn 1994): 228–266.

Mumford, Lewis. “Technics and the Nature of Man.” Technology and Culture 7, no. 3 (Summer 1966): 303–317.